

ABIEL J. KIM

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EDUCATION

Bachelor of Science | *Computing Science, Artificial Intelligence Specialization*
Simon Fraser University

May 2025
Burnaby, Canada

WORK EXPERIENCE

Machine Learning Engineer

February 2025 – September 2025

MICA Technologies Inc. | 

Vancouver, BC

- Achieved **production-grade** mIoU classification accuracies in an **accelerated** time frame
- Collaborated with geologists on **thin-section grain analyses** and inter-domain requirements
- Researched and developed proprietary architectures and **data processing** techniques
- Elicited client requirements to identify **workflow bottlenecks** and technical insights

Full Stack Software Developer

May 2022 – August 2022

3AG Systems | 

Burnaby, BC

- Implemented UI dashboards in **Angular** and developed **backend APIs** in Node.js
- Built **interactive data analytics panels** with backend APIs to enable dynamic data rendering
- Handled **backend services** with AWS cloud infrastructure using REST APIs
- Debugged client and server end bugs to **optimize performance** and satisfy UX requirements

Technical Project Manager, Product Engineer (Co-Op)

September 2021 – April 2022

Arlo Technologies Inc. | 

Richmond, BC

- Verified **firmware builds** and soldered **PCB circuits** of both internal and external products
- Assisted in **program oversight**, logistics operations, and delegation of international tasks
- Executed **campus-wide** product experiments and **numerical analyses** to make internal decisions
- Evaluated, debugged, and resolved **defect products** to discover **root causes** in FW and HW

PROJECTS

SATMAP-7 | *Zustand, Redis, Three.js, R3F, Drei, SGP4, GLSL, TypeScript, Tailwind*

Spring 2026

Independent Project |  

- Tracked thousands of satellites in real-time via precise **SGP4 propagation** on **CelesTrak GP TLE data**
- Rendered interactive point clouds via **R3F** over a **3D globe** with **GLSL day/night texture shaders**
- Architected a **triple-layer Redis caching pipeline** to **optimize API ingestion** and server performance
- Engineered **satellite search** with a smooth **camera interpolation animation** upon selection via **Drei**
- Designed both **mobile and desktop only HUD UIs** for navigation, statistics or **orbit path presentation**

OSSAD-9 | *Zod, Zustand, Recharts, nmap, Firebase, TypeScript, Tailwind*

Spring 2026 (current)

Independent Project |  

- Architected a **7-toolkit / 30+ sub-tool** dashboard with **WCAG, SOC 2, and RFC compliance**
- Implemented a colour-coded **network visualization library** from scratch for **LAN star topologies**
- Spawned **nmap** ping scan instances for a given **CIDR** with **SSE event streaming and logging**
- Traced **hop-by-hop network paths** with **TTL, IP, ASN, and RTT statistics** and **sparklines**
- Diagnosed **MITM attacks** via **ARP conflict detection** using **MAC-to-IP, OUI, and interface data**

Crystal Clear | *MongoDB, Mongoose, Google OAuth, Cloudinary, JavaScript*

Spring 2026 (current)

Independent Project |  

- Implemented a custom, real-time, **user messaging system** using **MongoDB** and Javascript
- Engineered a **comment section UI** and backend with **reply-threads** and comment-like capabilities
- Built an asynchronous **search engine** with server integration and **query-prediction** capabilities
- Developed and tested my own user-personalized **content-recommendation** algorithms and backend
- Optimized server overloads via **paginated data fetches** and **client-server synchronization**

YouTube Generative AI | *HuggingFace, Python, Scikit, Matplotlib* Spring 2025

SFU, CMPT 713 NLP | 

- Designated **team lead** who designed our replica of YouTube's summary **generative AI** model
- Finetuned custom **Meta BART** and **Google T5** text-to-text transformers using **Kaggle and Python**
- Projected test data onto a 768D **sentence-embedding** space to identify **overlapping clusters**
- Optimized the cluster distribution via predefined K, **WCSS/Elbow Method**, and **Silhouette Scores**
- Experimented on and evaluated our generative models via **ROUGE n-gram overlap**

Brain-Computer Interface | *OpenBCI, Tensorflow, Python* Summer 2021

Independent Project |  

- Extracted **EEG brain waves** using the mark-IV **OpenBCI** 8-channel headset and Cyton board kit
- Trained neural networks to identify **motor-action intent** in the form of EEG (electroencephalography)
- Applied signal **processing theory** to compute max frequency rate for FFT EEG data streams
- Architected and programmed a **game engine and game loop** for **real-time visual feedback**
- Integrated the headset interface and inference model into the game loop to enable operability
- Demonstrated the capability to “**telepathically**” play the legendary game “Flappy Bird”

PUBLICATIONS AND RESEARCH

A Special Case of Nonlinear Extrapolation under the NTK Winter 2025

arXiv:2512.15749 | 

- Pioneered a **theorem** which discovers when **NTK ReLU MLPs nonlinearly extrapolate** at the origin
- Proved 1 remark, 2 lemmas, and the main theorem by **pseudo-inversion** of the **NTK Gram matrix**
- Contributed nuance to the scientific community on how wide neural networks behave

A Data Scaling Law of a Manifold's Resolution Spring 2025

SFU, CMPT 419 *Special Topics in AI* | 

- Theorized a **neural scaling law** on the power-law of data scaling and neural network loss
- Earned an A+ with feedback from academics at the University of Illinois Urbana-Champaign
- Recognized for **exceeding expectations** and delivering one of the top submissions in Spring 2025

HONORS AND AWARDS

Dean's Honour Roll (3x) Spring 2021, Fall 2024 – Spring 2025

Recognition of Outstanding Academic Excellence at SFU

Regional Individual Writing Tournament April 2017

2nd Place – *World Scholar's Cup*

International / Regional Debate Tournament (4x) October 2016 – March 2017

2nd Place – *CAIMUN, VMUN, BMUN*, 3rd Place – *NVMUN*

Western Canadian Business Case Competition February 2017

Finalist – University of British Columbia *miniEnterprise*

LEADERSHIP AND INVOLVEMENT

Founder and President September 2015 – June 2018

Carson Graham Programming Club

Vice President of IT March 2017 – June 2017

Wish Youth Network Society

SKILLS

Languages: English (Native), Korean (Proficient), French (Limited)

Programming: Bash, C/C++, Java, React, JavaScript, Next, MATLAB, PHP, Python, TensorFlow, PyTorch, SQL

Documentation: LaTeX, Microsoft Office Suite, Markdown